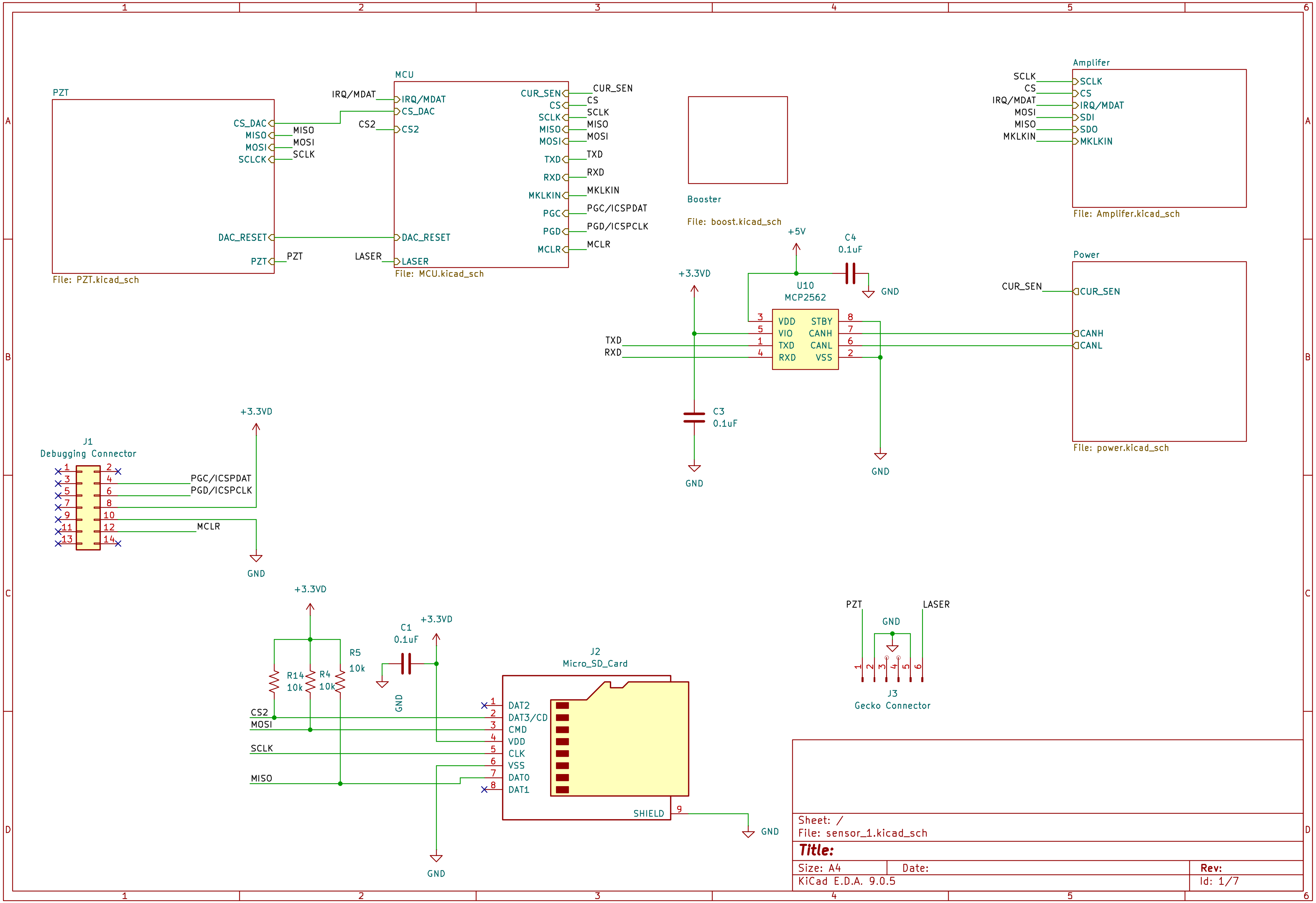
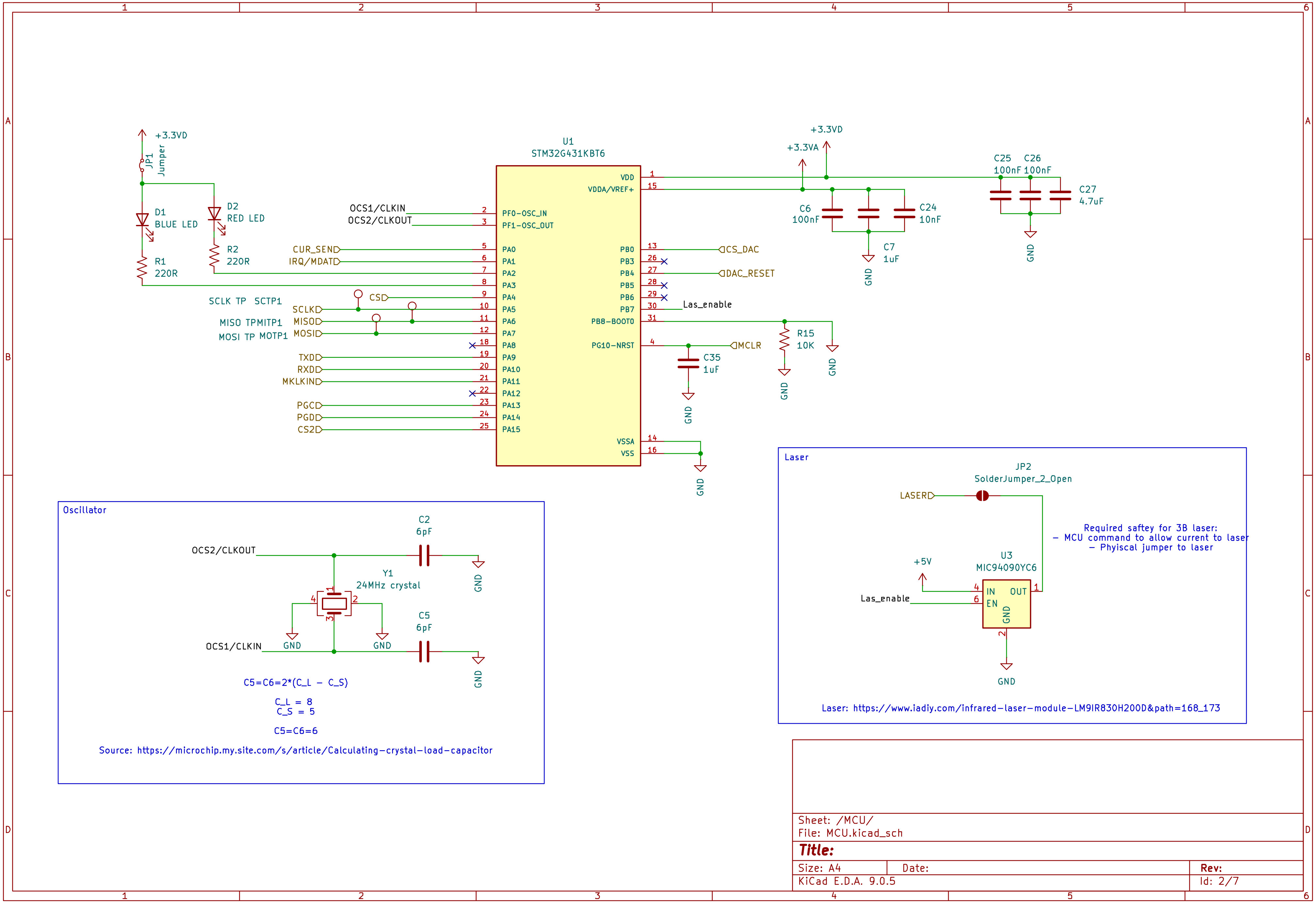
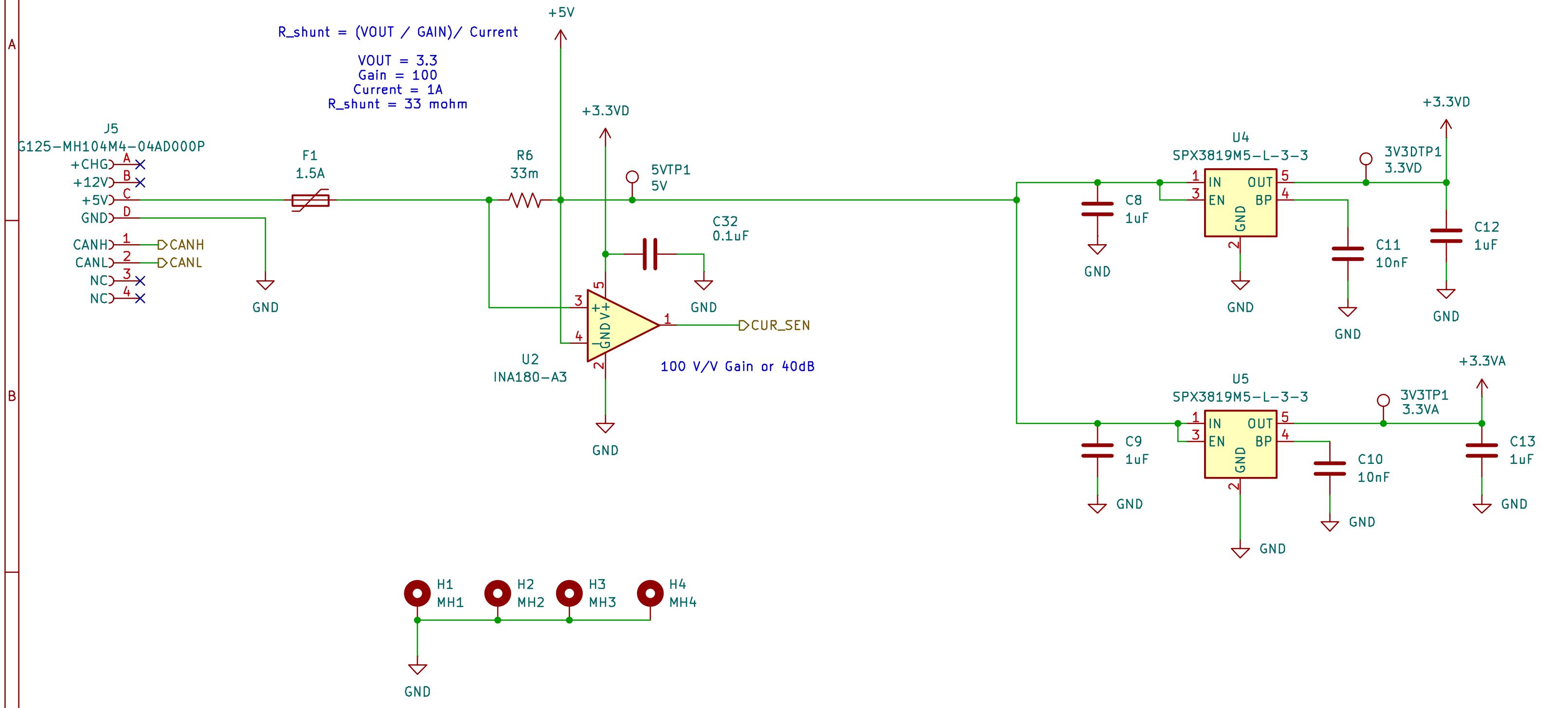

PZT
CS_DAC
MISO
MOSI
SCLCK
DAC_RESET
PZT
File: PZT.kicad_sch
MISO
MOSI
SCLK
IRQ/MDAT
CS2
LASER
MCU
IRQ/MDAT
CS_DAC
CS2
DAC_RESET
LASER
File: MCU.kicad_sch
CUR_SEN
CS
SCLK
MISO
MOSI
TXD
RXD
MKLKIN
PGC
PGD
MCLR
CUR_SEN
CS
SCLK
MISO
MOSI
TXD
RXD
MKLKIN
PGC/ICSPDAT
PGD/ICSPCLK
MCLR
Booster
File: boost.kicad_sch
Amplifer
SCLK
CS
IRQ/MDAT
MOSI
MISO
MKLKIN
SCLK
CS
IRQ/MDAT
SDI
SDO
MKLKIN
File: Amplifer.kicad_sch
+5V
C4
0.1uF
GND
+3.3VD
U10
MCP2562
VDD
VIO
TXD
RXD
STBY
CANH
CANL
VSS
TXD
RXD
C3
0.1uF
GND
GND
Power
CUR_SEN
CUR_SEN
CANH
CANL
File: power.kicad_sch
+3.3VD
J1
Debugging Connector
PGC/ICSPDAT
PGD/ICSPCLK
MCLR
GND
+3.3VD
R14
10k
R4
10k
R5
10k
C1
0.1uF
+3.3VD
GND
CS2
MOSI
SCLK
MISO
GND
J2
Micro_SD_Card
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
SHIELD
GND
PZT
LASER
GND
J3
Gecko Connector
Sheet: /
File: sensor_1.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.5
Id: 1/7

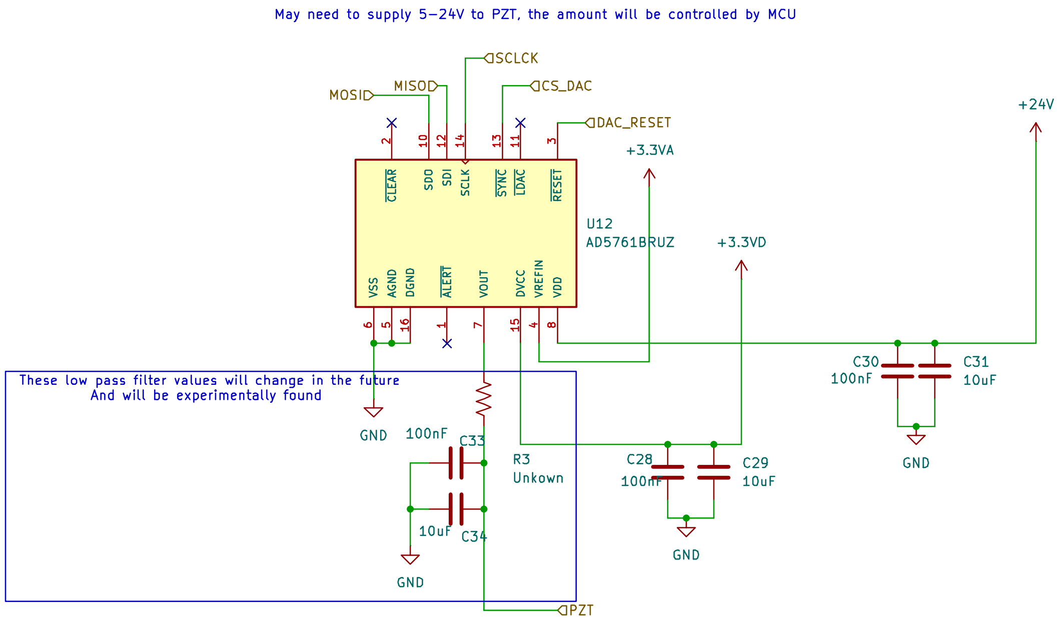

U1
STM32G431KBT6
VDD
VDDA/VREF+
PF0-OSC_IN
PF1-OSC_OUT
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB3
PB4
PB5
PB6
PB7
PB8-BOOT0
PG10-NRST
VSSA
VSS
+3.3VD
+3.3VA
JP1
Jumper
D1
BLUE LED
D2
RED LED
R1
220R
R2
220R
OCS1/CLKIN
OCS2/CLKOUT
CUR_SEND
IRQ/MDAT
CS
SCLK TP SCTP1
SCLK
MISO TPMITP1
MISO
MOSI TP MOTP1
MOSI
TXD
RXD
MKLKIN
PGC
PGD
CS2
CS_DAC
DAC_RESET
Las_enable
MCLR
R15
10K
C35
1uF
C6
100nF
C7
1uF
C24
10nF
C25
100nF
C26
100nF
C27
4.7uF
GND
Oscillator
C2
6pF
Y1
24MHz crystal
C5
6pF
C5=C6=2*(C_L - C_S)
C_L = 8
C_S = 5
C5=C6=6
Source: https://microchip.my.site.com/s/article/Calculating-crystal-load-capacitor
Laser
JP2
SolderJumper_2_Open
LASER
Required saftey for 3B laser:
- MCU command to allow current to laser
- Phyiscal jumper to laser
+5V
U3
MIC94090YC6
IN
EN
OUT
Laser: https://www.iadiy.com/infrared-laser-module-LM9IR830H200D&path=168_173
Sheet: /MCU/
File: MCU.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.5
Id: 2/7




Sheet: /Power/
File: power.kicad_sch

**Title:**

Size: A4 | Date: | Rev:
KiCad E.D.A. 9.0.5 | Id: 3/7




Acceptable Voltage Flucation of +-0.5V
Low pass introduced to force the acceptable voltage range

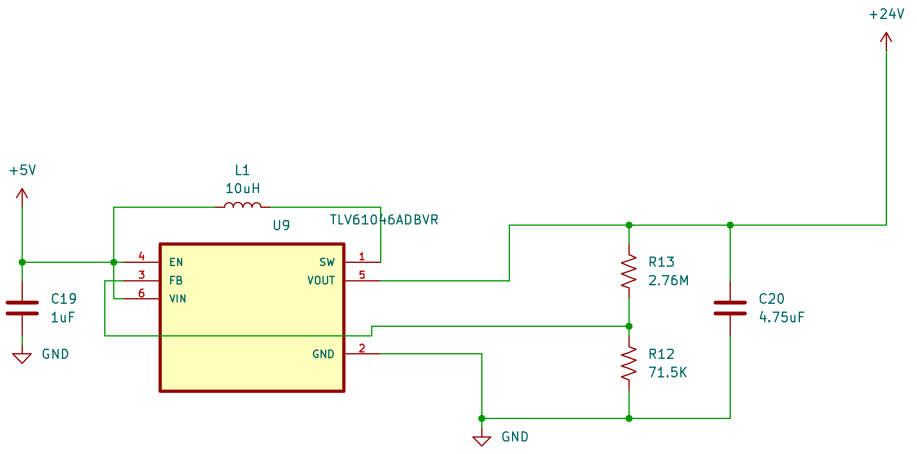
Reistior to be found experimentally

Proof in google sheets labbeled "Experimental & PCB Data Setup Master" under Photodiode Experimental Data,
Requirments & Comparison of PZT & Comparison of PZT to Generated Current in Payload GOogle drive

Sheet: /PZT/
File: PZT.kicad_sch

**Title:**

| | | |
|---|---|---|
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 9.0.5 | | Id: 4/7 |

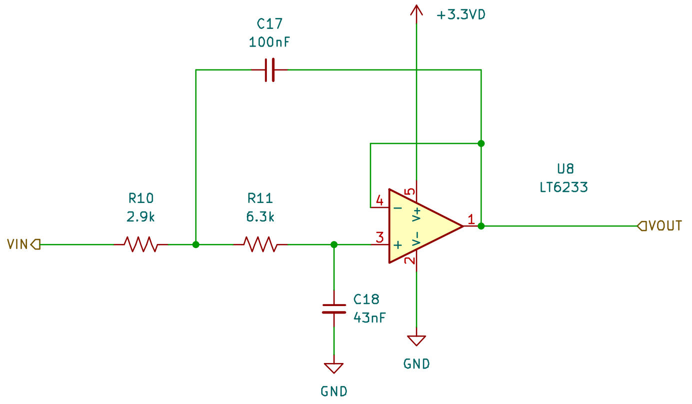



Reference photos, calculations all in Payload Experimental & PCB Data Setup Master
under Booster sheet

Sheet: /Booster/
File: boost.kicad_sch

**Title:**

| Size: A4 | Date: | Rev: |
|---|---|---|
| KiCad E.D.A. 9.0.5 | | Id: 4/7 |

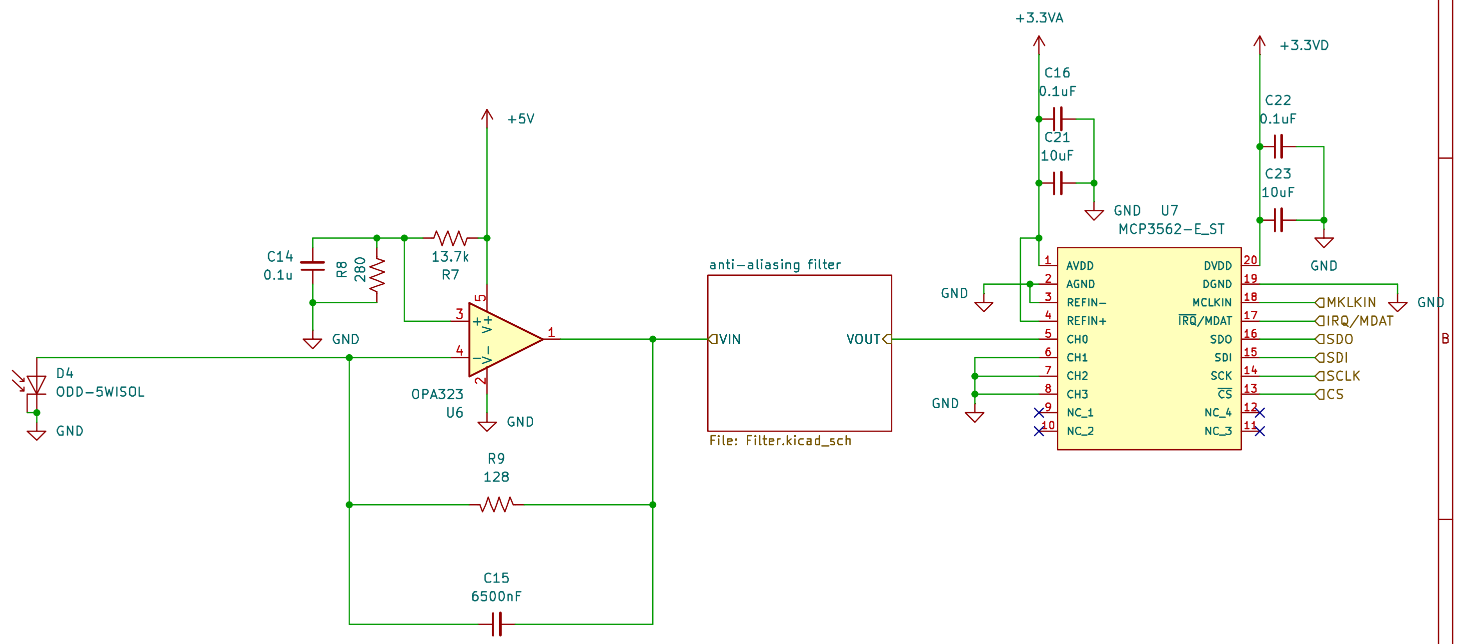



Math is done on a google sheets labbeled "Experimental & PCB Data Setup Master" under AA Filter
Google sheet is found under payload 2026 under data magnement

Sheet: /Amplifer/anti–aliasing filter/
File: Filter.kicad_sch

**Title:**

| Size: A4 | Date: | Rev: |
|---|---|---|
| KiCad E.D.A. 9.0.5 | | Id: 4/7 |




Math is done on a google sheets labbeled "Experimental & PCB Data Setup Master" under Transimpedance Amplifer
Google sheet is found under payload 2026 under data magnement

Sheet: /Amplifer/
File: Amplifer.kicad_sch

**Title:**

Size: A4 Date: Rev:
KiCad E.D.A. 9.0.5 Id: 5/7